Free Solo
Problem ID: freesolo

Alex is a free solo climber, meaning he likes to scale mountains without any safety gear. He just completed another risky climb, and is currently enjoying the view from the top.

However, due to the adrenaline rush during his ascent, he forgot to note the safest path back down. Knowing that climbing back down is more complicated, he wants to minimize the risk of falling to his death.

Since he has 20/20 vision, and due to the shape of the mountain, he can see all the different sections he can traverse on the way down. After each completed section, he can only continue his descent in sections directly below the one he just completed, either to the left or to the right.

Having plenty of experience with risky climbs, he is able to estimate how dangerous each section is. Unfortunately, he struggles to piece the sections together to find the least risky path overall.

Input

The first line contains an integer N $(1 \leq N \leq 1000)$, the number of rows in the mountain path.

Each of the next N lines contains a list of integers. The i-th of these lines contains exactly i integers, representing the risk values of the positions on the i-th row of the mountain. Each integer M is in the range $-1000 \leq M \leq 1000$.

Output

A single integer that represents the total risk value of the least dangerous path down.

Sample Input 1

```
4
2
3 4
6 5 7
4 1 8 3
```

Sample Output 1

```
11
```

Sample Input 2

```
4
2
3 4
1 6 5
1000 1000 2 1
```

Sample Output 2

```
12
```

Sample Input 3

```
1
100
```

Sample Output 3

```
100
```

Sample Input 4

```
4
4
-3 2
8 6 10
0 -1 10 -15
```

Sample Output 4

```
1
```